CS 391
Web Application Development

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Week - 8

Outline

1. What is Data Binding?
2. Passing an event handler to a component

What is Data Binding?

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2. Passing an event handler to a component

What is Data Binding?

From Wikipedia:
Data-binding is a technique that binds **data sources** from the **provider and consumer** together and synchronizes them.

The connection between the component's logic and the data it displays is called data-binding

Data-binding types

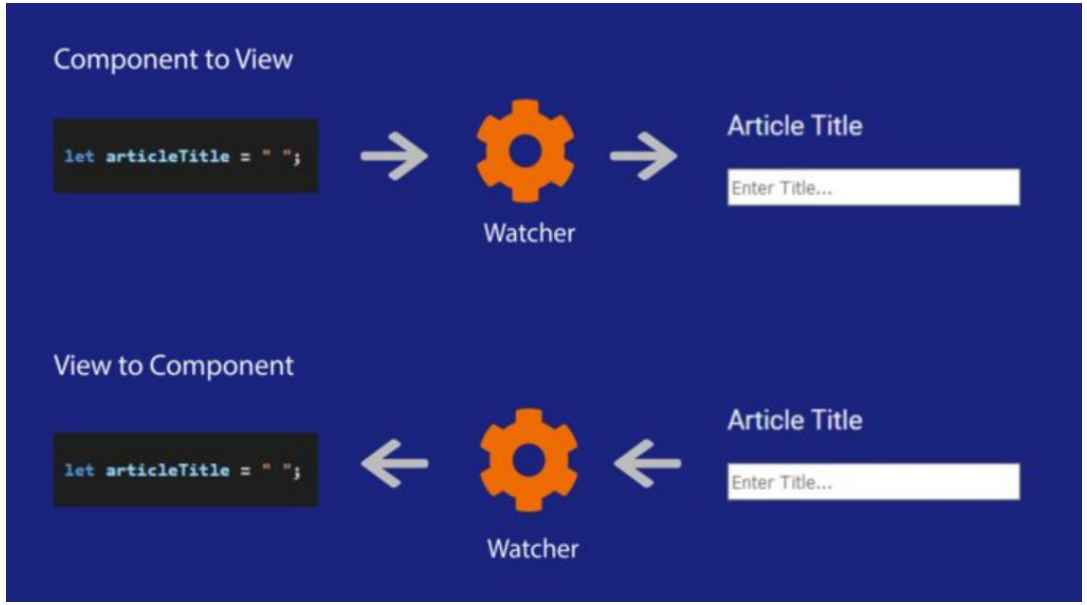
- There are two types of data-binding:
  - One-way data binding
  - Two-way data-binding

One-way data binding

- The data is bound one-way
- One-way data-binding implies that only one of the following connections can be applied:
  - Any change in the component's logic (data) gets reflected inside the UI (view).

    OR

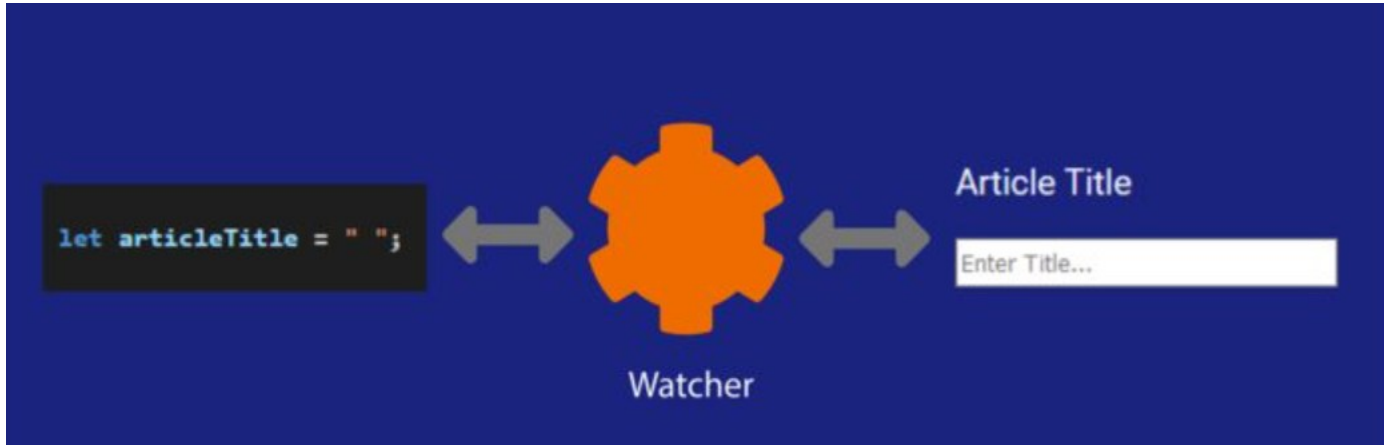
  - Any change in the UI (view) gets reflected inside the component's logic (model)



Two-way data-binding

- In two-way data-binding, the following connections are made at once:
  - Change in the view gets reflected in the component's logic.

    AND
  - Change in the component's logic gets reflected in the view.



any change in view reflects in the component's logic and any change in component's logic reflects in the view.

Data-binding in React

- By default, the connection between a component and a view is always **one-way** bound.
- React was never really designed for two-way binding although two-way data-binding can be implemented in React.
- We can apply additional logic to a component so that the connection is made two-way.

One-way data-binding in React

- In the case of React, we can only perform **components to view connection.**
- This means in React, **by default** there is no way that a view can change the component's logic **directly**
- Only the component's logic can affect the view.

View to Component

- Event handlers can be added to the view elements and change the component's data.

```
const Table = (props) => {
    const {products, onDelete}=props;
    return (
        <table> ...
        {products.map((item, index) => {
            return (
              <tr key={index}>
                 <td>{item.name}</td>
                ...
                 <td><button onClick={()=>onDelete(item.id)}>Delete</button></td>
                </tr>
            );
        })}
</table>
```

Passing an event handler to a component

1. What is Data Binding?
2. Passing an event handler to a component

Handling events with React elements…

- Handling events with React elements is very similar to handling events on DOM elements.
- There are some syntax differences:
  - React events are named using camelCase, rather than lowercase.
  - With JSX you pass a function as the event handler, rather than a string.

HTML

```
<button onclick="activateLasers()">
  Activate Lasers
</button>
```

JSX

```
<button onClick={activateLasers}>
  Activate Lasers
</button>
```

...Handling events with React elements

- You cannot return false to prevent default behavior in React.
- You must call preventDefault explicitly.

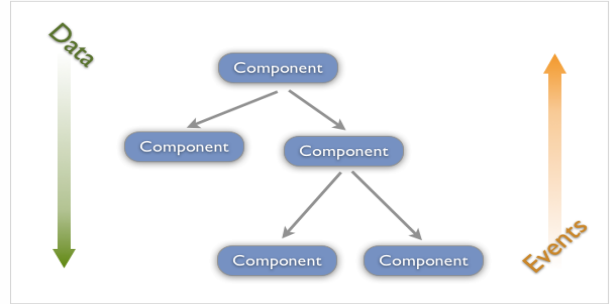
HTML

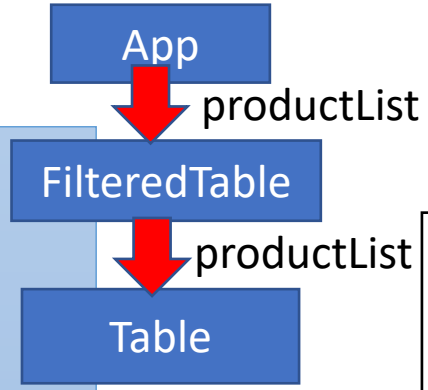
```
<a href="#" onclick="console.log('The link was clicked.'); return false">
  Click me
</a>
```

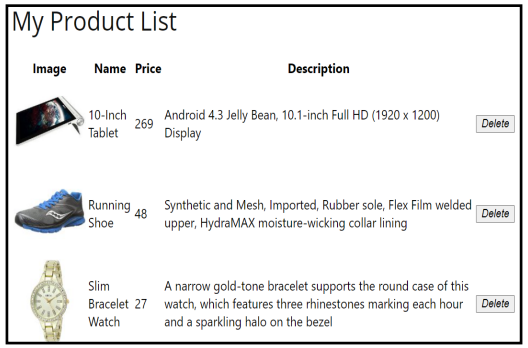
JSX

```
function ActionLink() {
    function handleClick(e) {
      e.preventDefault();
      console.log('The link was clicked.');
    }
    return (
      <a href="#" onClick={handleClick}>Click me</a>
    );}
```

Example








```
import { useState } from 'react';

import {data} from './data.js'

import FilteredTable from './components/FilteredTable';

function App() {
  const [productList, setProductList] = useState(data);

  const onDelete = (id) => {
    const filteredList =  productList.filter((product) => {
      return product.id !== id;
    });

    setProductList(filteredList);
  }
  return (...);
}

export default App;
```

```
return (
   <div className="App">
    <PageHeader/>
    <FilteredTable
      products={productList}
      onDelete={onDelete}>
    </FilteredTable>
   </div>
 );
```

Example

```
import Table from './Table’;
import FilterBar from './FilterBar';
const FilteredTable = (props) => {
    const { products, onDelete } = props;
    return (
      <div>
         <FilterBar></FilterBar>
         <Table products={products} onDelete={onDelete} ></Table>
      </div>
    );
}
export default FilteredTable;
```

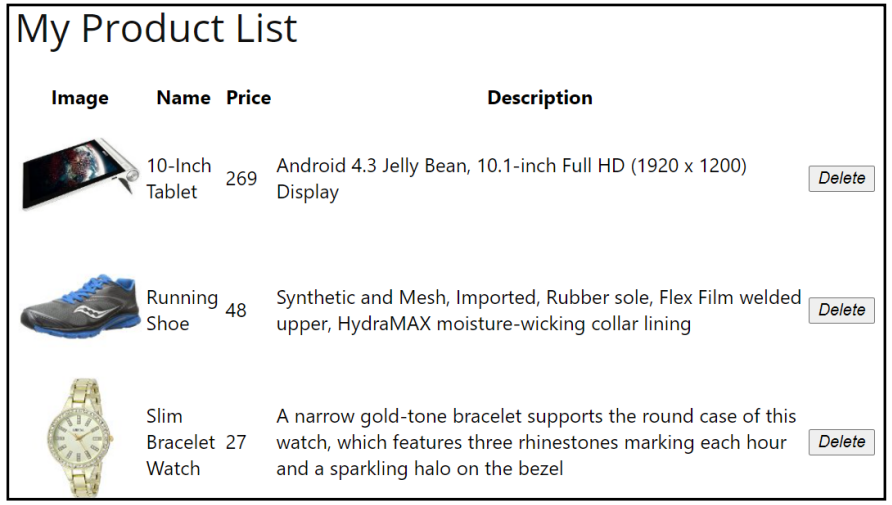
Example

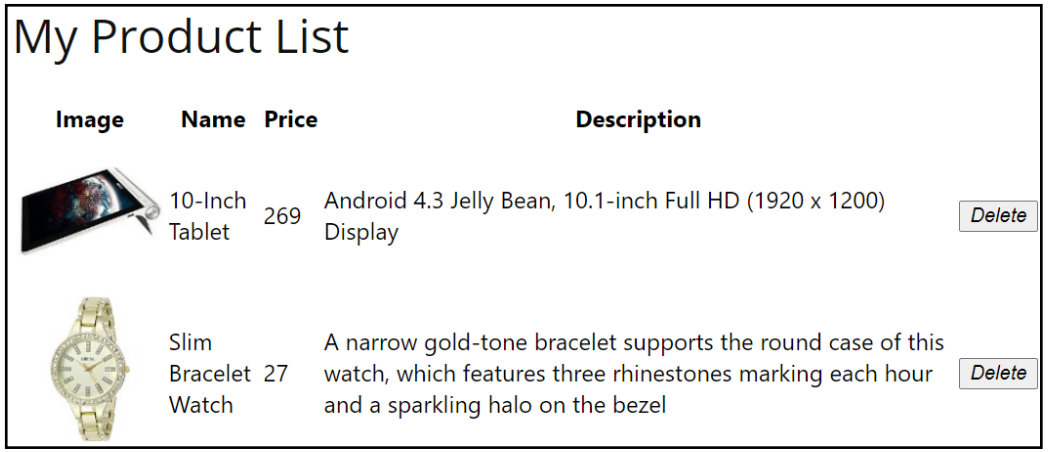
```
const Table = (props) => {
    const {products, onDelete}=props;
    return (
    <table>
        <tr><th>Name</th>
            <th>Price</th>
            <th>Description</th>
        </tr>
        {products.map((item, index) => {
            return (
                <tr key={index}>
                    <td>{item.name}</td>
                    <td>{item.price}</td>
                    <td>{item.description}</td>
                    <td><button onClick={()=>onDelete(item.id)}>Delete</button></td>
                </tr>
            );
        })}
    </table>
    );
}
export default Table;
```

In React, **event handlers expect a function**. If you call it directly like onDelete(item.id) → it executes immediately when the component renders. Using an arrow function wrapper (()=>...) → ensures it only executes **when the click event occurs** and allows you to pass parameters

Output

When you click ***Delete*** button it will remove the selected row from the table.





Any Question?

Thank you for your attention

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